



LESSON 4.4

PLOTTING, ORDERING, AND COMPARING RATIONAL NUMBERS – PART 1

LESSON INTRODUCTION

MA.6.NSO.1.1 Extend previous understanding of numbers to define rational numbers. Plot, order, and compare rational numbers.

LEARNING TARGETS

- I can plot, compare, and order fractional representations of rational numbers.

BENCHMARK COHERENCE

BUILDING ON

MA.5.NSO.1.4 Plot, order, and compare multidigit numbers with decimals up to the thousandths.

WORKING TOWARD

MA.8.NSO.1.1 Extend previous understanding of rational numbers to define irrational numbers within the real number system. Locate an approximate value of a numerical expression involving irrational numbers on a number line.

ESSENTIAL QUESTIONS

- How do you plot rational numbers on number lines?
- How do you compare rational numbers?

LESSON NARRATIVE

Students are introduced to plotting and comparing (using $<$, $>$, or $=$) rational numbers represented as fractions. Students participate in an activity where they sort sets of numbers building on Lesson 3, by including positive and negative decimals, fractions, and percentages.

COMPONENT SUMMARY

4.4.1 WARM-UP (~5 MIN.)

Determine which of two rational numbers is closer to zero

4.4.2 EXPLORATION (10 MIN.)

Explore how to plot and compare fractions

4.4.3 ACTIVITY (15-20 MIN.)

Order rational numbers written on cards

4.4.4 YOUR TURN! (10 MIN.)

Practice plotting, ordering, and comparing rational numbers

4.4.5 WRAP-UP (~5 MIN.)

Self-assessment of understanding of lesson's learning targets

RESOURCES

GLOSSARY

Key Terms:

- N/A

Additional Terms:

- N/A

MATERIALS

- Cards for 4.4.3 Activity

ADDITIONAL PREPARATION

- Consider preparing number lines with $\frac{1}{2}$ and/or $\frac{1}{4}$ benchmarks.
- Print and cut out cards for 4.4.3 Activity.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may struggle when comparing and ordering fractions if they do not recall how to rewrite fractions with common denominators.
- Students may struggle when comparing and ordering decimals if they do not recall how to line up the place values.

THINGS TO CONSIDER

- Consider creating a reference poster with instructions and examples of ordering and comparing fractions during 4.4.2 Exploration.

LITERACY AND LANGUAGE ROUTINES

Compare and Connect

4.4.2 Exploration and 4.4.3 Activity provide opportunities for students to discuss different strategies for comparing, ordering, and plotting rational numbers. Facilitate discussions to help students connect new strategies to prior knowledge.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.4.1 Discussion and Reflection

Students have opportunities to discuss and reflect on the strategy they used to determine which number is closer to zero in the 4.4.1 Warm-Up. Encourage and model revoicing and providing constructive criticism.

DIFFERENTIATE INSTRUCTION

A number line (labeled from -5 to 5) can be used as a scaffolding tool in 4.4.3 Activity. Including $\frac{1}{2}$ and $\frac{1}{4}$ benchmarks and using different colors on the number line will provide a visual entry point.

Students could also utilize digital number lines to verify their answers as they work.

4.4.1 WARM-UP: CLOSER TO ZERO

Component Overview: Students determine and then discuss which rational number is closest to zero from four given pairs of numbers.

TEACHER MOVES

Monitor for students who converted values to be in the same form (decimal or fraction) compared to students who compared in different forms. Elicit whole-group responses for students to compare their approaches. Students can also compare strategies like using a number line or estimating based on nearest whole numbers.



4.4.1 Warm-Up: Closer to Zero

- For each pair of numbers, determine which value is closest to 0.

Pair	Closest to Zero
$\frac{9}{11}$ or $\frac{15}{11}$	$\frac{9}{11}$
$\frac{1}{5}$ or $\frac{1}{9}$	$\frac{1}{9}$
1.25 or $\frac{5}{4}$	Neither. $1.25 = \frac{5}{4}$
0.01 or 0.001	0.001

- Discuss with your partner the strategy you used.



4.4.2 Exploration: Plotting and Comparing Fractions

1. Plot and label the rational numbers on the number line below.



Using the rational numbers plotted above, answer the following questions.

2. Andre says that $\frac{1}{4}$ is less than $-\frac{3}{4}$ because out of the two values, $\frac{1}{4}$ is closer to 0. Explain why you agree or disagree with Andre.

Answers will vary. I disagree because $-\frac{3}{4}$ is negative, so it is less than $\frac{1}{4}$ because all negative numbers are smaller than positive numbers.

3. Complete the table for each pair of numbers.

Pair	Farthest Distance From 0	Greater Value
$\frac{1}{4}$ or $\frac{5}{4}$	$\frac{5}{4}$	$\frac{5}{4}$
$-\frac{3}{4}$ or $\frac{5}{8}$	$\frac{3}{4}$	$\frac{5}{8}$

4. Is the value that is farther from 0 always the greater value? Explain your reasoning.
Answers will vary. No, the value that is farther from 0 is not always the greater value because it depends upon the sign of that value.

4.4.2 EXPLORATION: PLOTTING AND COMPARING FRACTIONS

Component Overview: Students explore how to plot and compare positive and negative fractions using a number line. Circulate while students work to monitor progress, assist struggling students, and select students to share their strategies during the whole-class debrief.

TEACHER MOVES

Compare and Connect

Prompt students to compare and connect their prior knowledge of plotting and comparing integers and positive fractions to plotting and comparing positive and negative fractions.

DIFFERENTIATE INSTRUCTION

Consider providing digital number lines for students to verify their answers as they explore.



4.4.3 Activity: Ordering Rational Number Cards

Your teacher will give you a set of number cards. Order them from least to greatest.

$-3\frac{1}{2}$	$-2\frac{89}{100}$	$-2\frac{1}{5}$	$-1\frac{9}{10}$	$\frac{17}{100}$	$\frac{3}{4}$	$1\frac{3}{5}$	$\frac{5}{2}$
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Your teacher will give you a second set of number cards. Add these values to the correct places in the first ordered set.

$-3\frac{1}{2}$	$-2\frac{89}{100}$	$-2\frac{1}{5}$	$-1\frac{9}{10}$	10%	$\frac{17}{100}$	42%	0.5
$\frac{3}{4}$	81%	0.987	1.2	150%	$1\frac{3}{5}$	2.014	$\frac{5}{2}$

1. In the first round of ordering, how did ordering negative numbers compare to ordering positive numbers?
Answers will vary. For ordering negative numbers, the numbers further from zero were smaller than those closer to zero. But for ordering positive numbers, the opposite was true.
2. What strategies helped you determine how to order the rational numbers represented as a decimal or a percent after you had already ordered the fractions?
Answers will vary. I converted all the numbers to a decimal or a percent to help me order the rest of the numbers.

4.4.3 ACTIVITY: ORDERING RATIONAL NUMBER CARDS

Component Overview: Students order rational numbers written on cards. Students order fractions with different denominators in Round 1 and add cards with decimals and percentages to their ordering in Round 2. Students share their strategies at the end of each round.

DIFFERENTIATE INSTRUCTION

A number line (labeled from -5 to 5) can be used as a scaffolding tool. Including $\frac{1}{2}$ and $\frac{1}{4}$ benchmarks and using different colors on the number line will provide a visual entry point.

ENRICHMENT

- How did you decide where to place specific numbers when ordering the numbers?
- Which numbers were the hardest to place? Which were the least difficult?
- How did you use the numbers from Round 1 to determine where to place new numbers in Round 2?

4.4.4 YOUR TURN!

Component Overview: Students practice plotting, ordering, and comparing rational numbers.

QUESTIONING

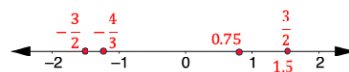
Prompt students to create their own real-world questions that require the ordering and/or comparing of rational numbers. Have students trade and solve each other's questions.



4.4.4 Your Turn!

1. Plot the following values on the number line. Label each point with its numeric value.

$$1.5 \quad -\frac{3}{2} \quad -\frac{4}{3} \quad 0.75 \quad \frac{3}{2}$$



2. Use the symbols $<$, $>$ or $=$ to compare each pair of values.

$$-\frac{4}{3} > -\frac{3}{2} \quad 0.75 < \frac{3}{2} \quad \frac{3}{2} = 1.5$$

3. Order the set of rational numbers.

$$5.2 \quad -3\frac{1}{2} \quad 5\frac{1}{2} \quad -2\frac{3}{4}$$

$$\text{Least to greatest: } -3\frac{1}{2} \quad -2\frac{3}{4} \quad 5.2 \quad 5\frac{1}{2}$$

$$\text{Greatest to least: } 5\frac{1}{2} \quad 5.2 \quad -2\frac{3}{4} \quad -3\frac{1}{2}$$

**4.4.5 Wrap-Up: Reflection**

1. Circle the emoji that best represents your understanding of the lesson today.
Answers will vary.



0 1 2 3 4 5 6 7 8 9 10

2. Write an explanation of why you chose that emoji.
Answers will vary.

4.4.5 WRAP-UP: REFLECTION

Component Overview: Students complete a self-assessment of their understanding of the lesson's learning targets using an emoji scale. Then students provide a written explanation to support their choice of emoji.

DIFFERENTIATE INSTRUCTION

Consider grouping students for collaboration in Lesson 5 based on their response. Alternatively, consider placing a number line on chart paper for students to mark as they walk out of the door for a visual spread.